SliderMC Command Cheat Sheet

Firmware V1.0
DIN A4 · ASCII protocol

S — Set (session)

SS	Set Speed	Cruise speed mm/s (≤ max_speed_1); bare reloads init_speed; live on next fill (incl. MJ). Dual MT: axis1=session, axis2×ratio.
SA	Set Accel	Peak accel [, decel] mm/s² (≤ max_accel_1). One value sets both ramps; two split start/stop. No skip _. Bare reloads init_accel into both; live on next fill (incl. MJ). Dual MT: same ratio scaling as SS.
SE	Set Enable	Driver enable 0|1; bare toggles; required before motion; off = hard stop. SE 0 also stops servo PWM (limp).
ST	Set Terminal	Terminal Mode 0|1; bare toggles; local echo + UART sniff to USB (expert).
SV	Set Verbose	Verbose #… push 0|1; bare toggles; default 10 Hz while moving (verbose_rate_hz). Idle is change-triggered plus a 1 s heartbeat.
SD	Set Debug	USB-only debug level 0..5; bare restores default; never on UIC UART.
SL	Set Left	Session soft min (working window); bare→MOTOR_N_min/SERVO envelope; none clears (→envelope if set); skip _; !E:limit past envelope.
SR	Set Right	Session soft max; bare→MOTOR_N_max/SERVO envelope; none clears; skip _; !E:limit if left>right.
SP	Set Position	Set reported pose (no motion); idle only; bare/0 = here is zero; skip _.

G — Get (session)

GS	Get Speed	Current session cruise speed.
GA	Get Accel	Current session accel and decel (mm/s²).
GE	Get Enable	Driver enable state.
GT	Get Terminal	Terminal Mode state.
GV	Get Verbose	Verbose push state.
GD	Get Debug	USB debug level.
GL	Get Left	Session soft min; effective (session else envelope); - if both None; extra pipe fields when packed≥2.
GR	Get Right	Session soft max; same effective / - rules as GL.

I — Is / Info

IM	Is Moving	Moving or settling on any active axis.
IH	Is Homing	Homing cycle active.
IL	Is Limit	At soft-limit position (axis1).
IE	Is Error	PIN_DRV_ERROR / EMO latched.
IP	Is Position	One pipe field per packed live channel (motors then servos; CG axis = sum).
IA	Is Axis	Packed live channels motors+servos (1..7). Same as CG axis. CS axis rejected.
IT	Is Target	Axis-1 seek target, or - if none / soft-stop.
IR	Is Ready	1 only if idle, not homing, enabled, and not waiting.
IW	Is Waiting	1 if any WT / WM / WH / WP / WC / WN wait is active.
ID	Is Diag	Motion diag: four per-motor FIFO underrun counts, peak STEP Hz, four fifo_min levels.
IC	Is Cause	Last chip reset: power|wdt|run|soft|debug|brownout|…
IG	Is GPIO	ASCII GP / name / desc (≤80 cols). Extra-axis rows only if that axis is live.

M — Movement

MT	Move To	Absolute user units; up to 7 packed channels (XYZ motors 1–3, ABC servos; motor 4 positional); skip _; needs SE; live-retarget. Dual MT: time-sync. Out-of-window = !E:soft (no clip).
MB	Move By	Relative move; same skip/named XYZABC rules as MT.
MJ	Move Joy	Joy speed % of SS, signed (− left / + right); omit named extra=0; 0=soft-stop; SS/SA live; clamp max_speed_N. Hold-to-jog: MJ ±100, MS on release.
MH	Move Home	Homing; motor 1 (default) through live motors; no-op if home_mode_N=0; cancel MS/ME.
MS	Move Soft Stop	Soft decelerate both axes; keeps enable; ends joy-mode; does not cancel waits. Dual: scaled accel kept.
ME	Move E-Stop, Halt	Immediate STEP abort; enable off; cancel waits and remaining ; chain.

P — Path

PC	Path Clear	Clear path buffer (count→0); !E:busy while PG active.
PD	Path Data	Append signed µm sample(s); up to 7 packed channels; skip _ →0; OK while PG (live stream).
PG	Path Go	Play buffer from sample 0; needs SE; !E:empty|busy|disabled. MS/ME ends path.
PN	Path Number	Samples in buffer; allowed during PG.
PS	Path Slice	Slice length µs (≥1000); bare→init_path_slice_us; !E:busy while PG.

W — Wait

WT	Wait Time	Delay then continue ; chain; bare→1 s; never !E:timeout.
WM	Wait Moving	Pause chain until move ends; optional timeout → !E:timeout, cancel rest of chain.
WH	Wait Homing	Pause until homing ends; timeout same as WM.
WP	Wait Pos	Wait until time-sync master pos reached/overstepped; idle→immediate; 2nd arg=timeout.
WC	Wait Cruise	Wait until cruise (status M) or idle; optional timeout → !E:timeout.
WN	Wait Not cruise	Wait until not cruise M; idle/A/B→immediate; timeout same as WM.

E — Extender

EO	Ext Out	Ext out n logical 0|1; bare EO1 toggles; glued EO11≡EO1 1; OK during EMO. EO5+ rejected.

Special

RB	Reboot	Soft MCU reset (no power cycle); EN off first. CS motors/servos do not need RB.
BE	Beep	Pulse PIN_BUZZER; bare 100 ms; clamp 1..1000; not a wait. No-op if BUZZER_use=0. OK during EMO/path.
CT	Camera Trigger	Pulse PIN_CAMERA_CTRL low (OC sink); bare 100 ms; clamp 1..60000; not a wait; motion continues. OK during EMO/path. Verbose on: status letter T while pulsing (at least once).
HL/?	Help	Two-column table of all commands (short + phrase). ? needs newline; not realtime status.

C — Config

CS	Config Set	Persist key to mc.ini; silent ok. CS motors/servos re-init GPIO immediately (no RB). CS axis rejected.
CR	Config Reset	Reset all config to compiled defaults and save mc.ini.
CG	Config Get	One key, or bare dumps all keys (multi-line).

V — Version

VA	Version About	About string (name, version, author).
VF	Version FW	Firmware version.
VP	Version Protocol	Protocol version (1). Wire may change without bumping VP.
VG	Version GPIO	Machine-readable pin map. Extra-axis pins if that axis is live.

Now Commands

#	Status now	Realtime compact status line (no newline).
!	Soft Stop now	Realtime soft stop (same urgency as MS).
ESC	E-Stop now	Realtime halt (same urgency as ME): STEP abort, EN off, cancel waits/chain.

Wire: one command per line (`\n`); chain with `;`; axis args positional or named `X/Y/Z` (glued `MTX20Y50Z100` ok); `#` is realtime status, not a comment; `/` comments to end of line; bare bool setters toggle; motion/settings silent on success; errors `!E:code message`.
Realtime (no newline): `#` status · `!` soft stop · `ESC` halt (same as ME) · `Ctrl-X` (0x18) soft reset. `?` is help (needs newline), not status.
Status (`#X …`): `I` idle · `A` accel · `M` cruise · `B` decel · `H` homing · `P` path · `L` hard-limit · `D` disabled · `E` error · `T` camera trigger (CT pulse, or one-shot listen). Moving: `#M/#A/#B pos speed accel [target]`.
Halt vs Stop: `MS/!` soft decel (enable kept, ends joy-mode); `ME/ESC` immediate abort, enable off, cancel waits. `MJ`: skip if value unchanged. Hold-to-jog: `MJ ±100`, `MS` on release. `VP` stays 1 (wire may still change).